

# ALICJA NOWACKA

Postdoctoral Scientist - Immunology  
In vitro Models & Flow Cytometry



## PROFILE

- Immunologist with 8+ years of experience in complement and dendritic cell biology, primary immune cell isolation, and multicolour flow cytometry.
- Strong background in in vitro immune cell systems (DC-T cell co-cultures, functional assays) and in vivo mouse models (FELASA B, IACUC-aligned training).
- First-author and co-author of peer-reviewed publications on complement-mediated immune regulation, with multiple oral presentations and travel/poster awards at international conferences.
- Experienced in international collaborations (Germany, Sweden, USA), clear scientific communication, and cross-cultural teamwork; native in Polish, fluent in English, intermediate German and Swedish

## CONTACT

☎ +1 513 238 0125

✉ [alnowacka@gmail.com](mailto:alnowacka@gmail.com)

in <https://www.linkedin.com/in/alicia-nowacka-008b09262/>

id 0000-0003-2307-5948

📍 Lübeck, Germany

## EDUCATION

### PhD in Immunology

University of Lübeck, Germany  
IRTG1911 Program, Institute for Systemic Inflammation Research  
2020 - 2025  
Defense expected March/April 2026

### Master in Biomedical Chemistry

University of Gdańsk, Poland  
Department of Molecular Biotechnology  
2017 - 2020

### Bachelor in Biomedical Chemistry

University of Gdańsk, Poland  
Department of Molecular Biotechnology  
2014 - 2017

## SKILLS

### • Reliability & Integrity.

Careful, reproducible experimental work and transparent data reporting

### • Passion & Motivation

Strong drive to understand disease mechanisms and improve patient outcomes through translational research

### • Ownership & Responsibility

Takes full ownership of projects from design to analysis and presentation; manages priorities independently and meets deadlines

## PROFESSIONAL EXPERIENCE

### ○ Research Scientist / PhD Candidate, Immunology

Institute for Systemic Inflammation Research,  
University of Lübeck, Germany  
11/2020 - 09/2025

- Designed and executed in vivo and in vitro studies on anaphylatoxin-mediated regulation of conventional dendritic cell differentiation and function.
- Established and optimised multicolour flow cytometry panels for murine DC and T cell phenotyping; performed advanced data analysis using FlowJo and GraphPad Prism.
- Set up co-culture systems of DCs and T cells to dissect how complement receptors and TLR signals control T cell proliferation and differentiation.
- Isolated and cultured primary murine immune cells from bone marrow, spleen, blood, lung and brain, including functional readouts such as cytokine secretion and proliferation
- Coordinated data generation and exchange with international collaborators in Cincinnati; presented results at European and international complement meetings (multiple oral presentations and travel awards).

### ○ Visiting Researcher (Part of PhD Program)

Division of Immunobiology,  
Cincinnati Children's Hospital Medical Center, USA  
09/2022 - 02/2023

- Investigated the role of C5aR1 in antigen- and TLR-driven T cell proliferation and differentiation by splenic conventional dendritic cells.
- Isolated and analysed immune cells from mucosal tissues (e.g. lung) to study local immune responses and tissue-specific phenotypes.
- Applied multicolour flow cytometry and functional assays to characterise T cell responses in murine models.

### ○ Research Intern

Department of Translational Medicine, Lund University, Sweden  
09/2017 - 10/2018 & 09/2019 - 07/2020

- Contributed to projects on intracellular C3 in autophagy and pathogen detection, and on C3/C3aR function in human B cells.
- Performed molecular cloning, protein expression and purification, and flow cytometry-based assays under the supervision of Prof. Anna Blom and colleagues.

Additional research internships at the University of Gdańsk (molecular biotechnology, protein biochemistry, plant biotechnology) and local institutes (veterinary hygiene, biotech), details available upon request

## AWARDS & GRANTS

**ECN Poster Prize 2024**, 19th European Meeting on Complement in Human Disease (Lübeck, Germany).

**ICW 2023 Travel Award**, 29th International Complement Workshop (Newcastle, England).

**EMCHD 2022 Travel Award**, 18th European Meeting on Complement in Human Disease (Bern, Switzerland) for oral presentation on anaphylatoxin receptors and DC differentiation.

**Scholarship** from Lund University for students, **Academic Merit** and “**Best Students** of the University of Gdańsk, Faculty of Chemistry”.

## ADDITIONAL ACTIVITIES

Active in the **Board of European Students of Technology (BEST)** in Lund and Gdańsk, with roles in leadership, PR, and organisation of international summer courses and competitions; developed skills in project management, teamwork and science communication.

Member of **Boost Biotech** and university committees (student conference and grant funding, faculty board), supporting scientific events and student representation.

## LANGUAGES

- Polish – native
- English – advanced (written and spoken)
- German – intermediate
- Swedish – intermediate

## PUBLICATIONS

- **Nowacka A. A.**, Einwohlt P., Ohms M., Geisler S., Abdala Y., Fähnrich A., Köhl J. C3a and C5a regulate DC progenitor homeostasis and their differentiation into cDC1 and cDC2 subsets. Manuscript in preparation.
- **Nowacka A. A.**, Sordo Vieira L., Petr V., Fageräng B., Würzner R., Ohms M. The grand escape – how pathogens outsmart the human complement system. *Immunobiol.* 2025 230:153126
- Laumonnier Y., Korkmaz R. U., **Nowacka A. A.**, Köhl J. Complement-mediated immune mechanisms in allergy. *Eur. J. Immunol.* 2023 53: 2249979
- Kremlitzka, M., Colineau L., **Nowacka A. A.**, Mohlin F. C., Wozniak K., Blom A. M., King B. C. Alternative translation and retrotranslocation of cytosolic C3 that detects cytoinvasive bacteria. *Cell. Mol. Life Sci.* 2022 79: 291
- Kremlitzka, M., **Nowacka A. A.**, Mohlin F. C., Bompada P., Marinis Y. D., Blom A. M. Interaction of Serum-Derived and Internalized C3 with DNA in Human B Cells - A Potential Involvement in Regulation of Gene Transcription. *Front. Immunol.* 2019 10:493

## TECHNICAL SKILLS

### • Immunology & Cell Biology

Primary murine immune cell isolation (bone marrow, spleen, blood, lung, brain), in vitro DC differentiation, DC–T cell co-culture systems, functional assays (proliferation, cytokines), ELISA.

### • Flow Cytometry & Imaging

Multicolour flow cytometry panel design and analysis, experience with cell sorting as a user, immunohistochemistry, fluorescence microscopy.

### • In vivo Models

FELASA B certified; IACUC-aligned CITI training; mouse handling, intraperitoneal injections, work in **preclinical** disease and inflammation models.

### • In vitro Models

Experience with complex in vitro systems (immune cell co-cultures; interest in 3D/ALI tissue models and immune-epithelial cross-talk).

### • Molecular & Protein Biology

RNA extraction, cDNA preparation, RT-qPCR, DNA isolation, cloning, bacterial expression of recombinant proteins, protein purification, SDS-PAGE, Western blot, chromatography (affinity, ion-exchange, size-exclusion), TLC

### • Data & Software

GraphPad Prism, FlowJo, MS Office; figure and poster preparation with Adobe Illustrator, Photoshop and Affinity tools

## CONFERENCE HIGHLIGHTS

Oral and poster presentations at international meetings including EMCHD (Bern 2022, Lübeck 2024), ICW (Newcastle 2023) and AKKS (Germany, 2021–2025); multiple awards for scientific contribution and communication.

## REFERENCE

### Jörg Köhl

Prof. Dr. med. / Director of ISEF,  
Univ. of Lübeck, Germany

Phone +49 451 3101-8900

Email: [joerg.koehl@uksh.de](mailto:joerg.koehl@uksh.de)

### Chandrashekhar Pasare

Prof. / Director of Immunobiology,  
CCHMC, Cincinnati, USA

Phone: +1 513-803-3345

Email: [Chandrashekhar.Pasare@cchmc.org](mailto:Chandrashekhar.Pasare@cchmc.org)